



EAST WEST UNIVERSITY

Assignment

DSA 5005: Multivariate Statistics

Fall 2025

Submitted To:
Md. Sohel Rana (DSR)
Professor

Submitted By:
Rizvee Hassan Prito
ID: 2024-3-83-002
Section: 1

Department of Mathematical and Physical Sciences

1 No. Q. Ans.

Code:

```
data <- read.csv(file.choose())
head(data)
str(data)
data_new <- data.frame(data[, -length(data)])
colnames(data_new) <- c("CF", "SL", "SU", "FU", "FS", "DN", "Age", "CV")
head(data_new)
```

Output:

```
'data.frame': 3150 obs. of 9 variables:
 $ Call..Failure      : int  8 0 10 10 3 11 4 13 7 7 ...
 $ Subscription.Length: int  38 39 37 38 38 38 38 37 38 38 ...
 $ Seconds.of.Use     : int  4370 318 2453 4198 2393 3775 2360 9115 13773 4515 ...
 $ Frequency.of.use    : int  71 5 60 66 58 82 39 121 169 83 ...
 $ Frequency.of.SMS    : int  5 7 359 1 2 32 285 144 0 2 ...
 $ Distinct.Called.Numbers: int  17 4 24 35 33 28 18 43 44 25 ...
 $ Age                : int  30 25 30 15 15 30 30 30 30 30 ...
 $ Customer.Value      : num  198 46 1537 240 146 ...
 $ Churn               : int  0 0 0 0 0 0 0 0 0 0 ...
```

	CF	SL	SU	FU	FS	DN	Age	CV
1	8	38	4370	71	5	17	30	197.640
2	0	39	318	5	7	4	25	46.035
3	10	37	2453	60	359	24	30	1536.520
4	10	38	4198	66	1	35	15	240.020
5	3	38	2393	58	2	33	15	145.805
6	11	38	3775	82	32	28	30	282.280

Comment:

The dataset consists of 3,150 customer records across nine columns (variables). The last column, Churn, is the target variable and is excluded from the PCA analysis because PCA is an unsupervised technique. The variables Call Failures, Subscription Length, Seconds of Use, Frequency of Use, Frequency of SMS, Distinct Called Numbers, and Customer Value are renamed as CF, SL, SU, FU, FS, DN, and CV, respectively, to present the analysis output in a more structured and concise form.

Code:

```
s<- cov(data_new)
```

S

```
r <- cor(data_new)
```

r

Output:

```
> s<- cov(data_new)
> s
```

	CF	SL	SU	FU	FS	DN	Age	CV
CF	52.764034	10.5706930	1.529555e+04	239.11174	-18.16640	63.039865	2.6786725	455.2672
SL	10.570693	73.5045947	4.485075e+03	52.43941	73.43805	13.581677	-0.1839601	485.6592
SU	15295.554064	4485.0750467	1.762244e+07	228118.88060	48116.66901	48897.886234	772.6895200	900898.5393
FU	239.111738	52.4394081	2.281189e+05	3296.28791	644.51673	727.651388	-14.3739907	11920.1028
FS	-18.166400	73.4380472	4.811667e+04	644.51673	12597.26981	153.919047	-91.9800335	53669.2964
DN	63.039865	13.5816770	4.889789e+04	727.65139	153.91905	296.436708	7.7600982	2535.2501
Age	2.678672	-0.1839601	7.726895e+02	-14.37399	-91.98003	7.760098	77.9882477	-1006.1303
CV	455.267246	485.6592069	9.008985e+05	11920.10279	53669.29642	2535.250094	-1006.1302939	267304.9578

```
> r <- cor(data_new)
> r
```

	CF	SL	SU	FU	FS	DN	Age	CV
CF	1.00000000	0.169737274	0.50160659	0.57334936	-0.02228238	0.50405774	0.041757649	0.1212255
SL	0.16973727	1.00000000	0.12461763	0.10653394	0.07631777	0.09200897	-0.002429696	0.1095648
SU	0.50160659	0.124617629	1.00000000	0.94648923	0.10212319	0.67653630	0.020842867	0.4150872
FU	0.57334936	0.106533935	0.94648923	1.00000000	0.10001921	0.73611353	-0.028349814	0.4015725
FS	-0.02228238	0.076317768	0.10212319	0.10001921	1.00000000	0.07965044	-0.092798459	0.9248774
DN	0.50405774	0.092008969	0.67653630	0.73611353	0.07965044	1.00000000	0.051037177	0.2848074
Age	0.04175765	-0.002429696	0.02084287	-0.02834981	-0.09279846	0.05103718	1.00000000	-0.2203617
CV	0.12122548	0.109564756	0.41508724	0.40157246	0.92487743	0.28480743	-0.220361712	1.0000000

Comment:

The covariance matrix and correlation matrix reveal varying degrees of covariance and correlation among the variables, confirming the need for dimensionality reduction. A strong correlation exists between Frequency of Use and Seconds of Use (0.946), along with a strong covariance between them (2.28e+05), which serves as a clear example of highly related variables in the dataset. Additionally, a moderate correlation is observed between Customer Value and Seconds of Use (0.415), while a strong covariance exists between these variables (9.08e+05). The presence of different levels of correlation suggests that a smaller number of latent dimensions may effectively summarize customer behavior. Furthermore, the varying levels of covariance indicate strong linear relationships among several variables, suggesting significant redundancy in the feature set, as multiple variables capture overlapping information.

Code:

```
evl.evc <- eigen(s)
lambda <- evl.evc$values
lambda
e <- evl.evc$vectors
e
```

```
sum(diag(S))
```

```
sum(lambda)
```

Output:

```
> ev1.evc <- eigen(S)
>
> # Eigenvalues
> lambda <- ev1.evc$values
> lambda
[1] 1.767233e+07 2.325396e+05 6.264865e+02 3.856555e+02 1.198157e+02 7.315586e+01 3.501791e+01 2.983442e+01
>
> # Eigenvectors
> e <- ev1.evc$vectors
> e
      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]      [,7]      [,8]
[1,] -8.657880e-04 -1.469268e-03 -0.015213109 -0.136769107 0.022241360 0.1797412107 0.426441693 0.8754460382
[2,] -2.549030e-04 1.119265e-03 -0.004951721 0.011743914 0.071309966 0.9807196938 -0.090358716 -0.1574017686
[3,] -9.985703e-01 -5.102371e-02 -0.010284692 0.011605319 0.001812849 -0.0003862765 0.003034552 -0.0008837561
[4,] -1.292732e-02 4.412955e-04 0.144581349 -0.897452616 -0.393873436 0.0175119564 -0.109481085 -0.0779657938
[5,] -2.878391e-03 2.262025e-01 -0.931152466 -0.129901581 -0.075700779 -0.0005276671 0.203538845 -0.1332135851
[6,] -2.770987e-03 2.665584e-05 0.001138602 -0.397187145 0.910171264 -0.0722410557 0.033324003 -0.0865586934
[7,] -4.069296e-05 -4.469280e-03 -0.256080616 -0.002688857 0.069353213 -0.0188519361 -0.867705334 0.4199020143
[8,] -5.170500e-02 9.727309e-01 0.214734734 0.031002125 0.018122771 -0.0008471038 -0.050362398 0.0344020793
>
>
> # Total variance check
> sum(diag(S))
[1] 17906137
> sum(lambda)
[1] 17906137
> |
```

Comment:

The sum of eigenvalues equals the total variance, verifying correct decomposition.

Code:

```
pca <- prcomp(data_new, scale = TRUE)
```

```
pca
```

```
summary(pca)
```

Output:

```

> pca <- prcomp(data_new, scale = TRUE)
> pca
Standard deviations (1, ..., p=8):
[1] 1.8147577 1.3376420 0.9941045 0.9753448 0.7432519 0.6079201 0.2140944 0.1000013

Rotation (n x k) = (8 x 8):
      PC1      PC2      PC3      PC4      PC5      PC6      PC7      PC8
CF -0.35752013 -0.27317518 -0.12006797 0.094894545 -0.87038008 0.10338633 0.07481447 0.016910500
SL -0.11779082 0.01882711 -0.93312855 0.274517477 0.19547702 -0.03111647 -0.02234723 -0.001793875
SU -0.49593328 -0.14673183 0.09848040 -0.009286329 0.33149950 0.41427071 0.60846780 -0.266439711
FU -0.51017009 -0.16260569 0.12752004 0.041516475 0.21413897 0.26500332 -0.76060977 0.026294777
FS -0.19881852 0.66000683 -0.06697726 -0.255652134 -0.17601322 -0.07128521 -0.12924296 -0.634233584
DN -0.44205443 -0.19127856 0.09028387 -0.051645778 0.12925103 -0.85600960 0.08762797 -0.006708127
Age 0.02641034 -0.26412734 -0.27574876 -0.914472331 0.01885875 0.07106522 -0.02365352 0.106192349
CV -0.34123731 0.57886982 0.01485006 -0.097674878 -0.02092142 0.06009792 0.14213101 0.717255344

> summary(pca)
Importance of components:
      PC1      PC2      PC3      PC4      PC5      PC6      PC7      PC8
Standard deviation 1.8148 1.3376 0.9941 0.9753 0.74325 0.6079 0.21409 0.10000
Proportion of Variance 0.4117 0.2237 0.1235 0.1189 0.06905 0.0462 0.00573 0.00125
Cumulative Proportion 0.4117 0.6353 0.7589 0.8778 0.94682 0.9930 0.99875 1.00000

```

Comment:

Based on the cumulative proportion of variance, 87.78% of the total variance is explained by the first four principal components. The standard deviations indicate that PC1 explains the largest proportion of the total variance, followed by PC2, which explains the second largest proportion. The dominance of the first four components suggests strong correlations among the original variables, indicating that the dataset can be effectively represented in a lower-dimensional space.

Estimation of first two principal components scores:

PC1 = -0.357(Call Failures) - 0.117(Subscription Length) - 0.495(Seconds of Use) - 0.510(Frequency of Use) - 0.198(Frequency of SMS) - 0.442(Distinct Called Numbers) + 0.026 Age - 0.341(Customer Value)

PC2 = -0.273(Call Failures) + 0.018(Subscription Length) - 0.146(Seconds of Use) - 0.162(Frequency of Use) + 0.660(Frequency of SMS) - 0.191(Distinct Called Numbers) - 0.264 Age + 0.578(Customer Value)

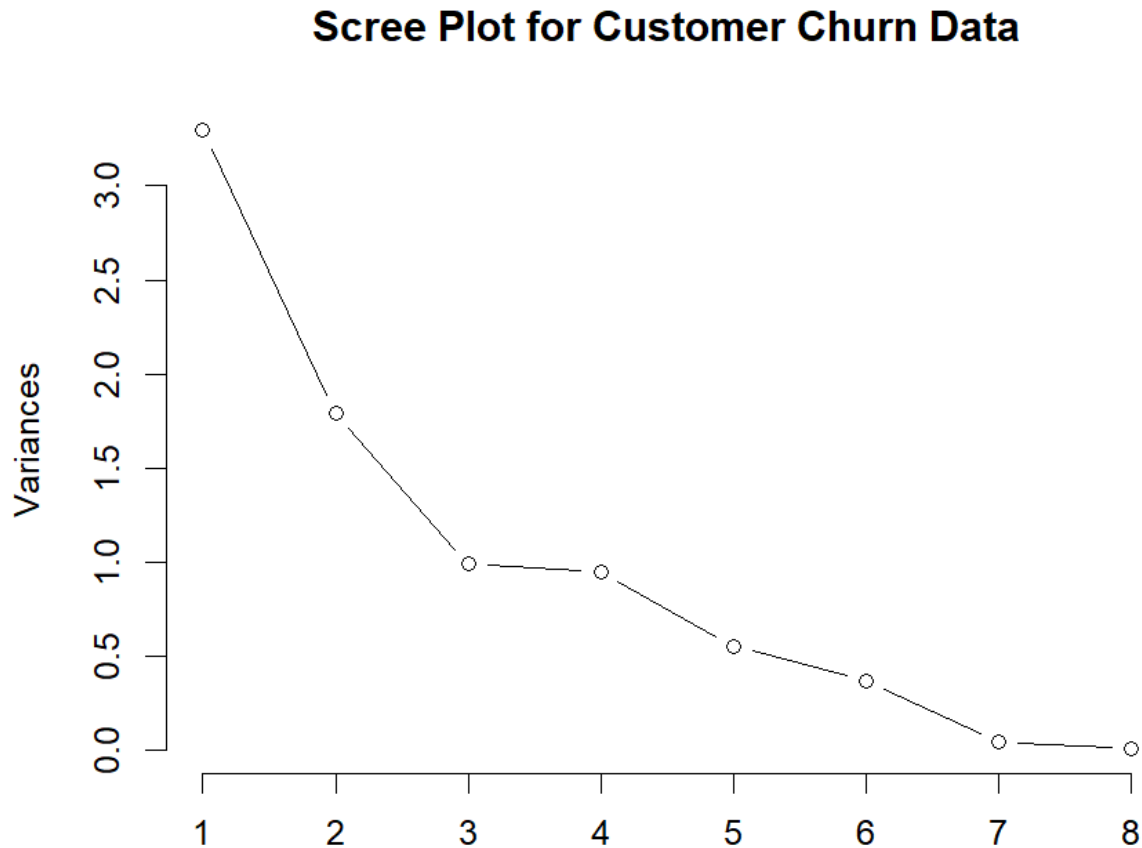
The first principal component is a combination of several call and usage variables, like Seconds of Use, Frequency of Use, Distinct Called Numbers, Call Failures, and Customer Value. We can think of this component as showing overall customer engagement with the telecom services. Customers who score high on this component use the services a lot and are very active, while customers with low scores are lighter users.

The second principal component shows a difference between customers who use SMS and generate more value and those who use more traditional voice services or are influenced by age-related behavior. This component can be seen as a communication style and value indicator, separating people who rely more on messaging and are high-value customers from those who use SMS less.

Code:

```
plot(pca, type = "l", main = "Scree Plot for Customer Churn Data")
```

Output:



This Figure shows a scree plot with eight principal components. An elbow occurs in the plot at about $i = 4$ or 5 . In this case, it appears that 4(or perhaps 5) sample principal components effectively summarize the total sample variance.

Overall Answer:

Principal Component Analysis (PCA) can be applied to the customer churn dataset to reduce the number of variables and reveal patterns in customer behavior. PCA works by transforming the original correlated variables into a smaller set of uncorrelated **principal components**, which capture most of the information in the data. In this dataset, the first four components explain about **87.78% of the total variance**, showing that customer behavior can be effectively summarized using fewer dimensions.

The first principal component (PC1) reflects overall customer engagement, combining call and usage variables to separate highly active customers from lighter users. The second principal component (PC2) captures differences in communication style and customer value, distinguishing high-value SMS users from customers who rely more on voice services or whose behavior is influenced by age. By focusing on these components, PCA reduces redundancy, simplifies the dataset, and highlights important patterns in usage, engagement, and customer value.

2 No. Q. Ans.

Code:

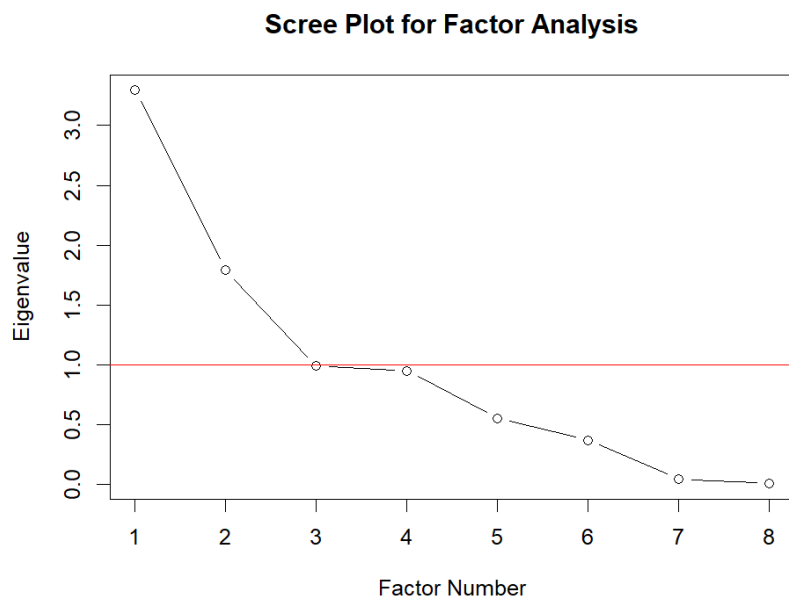
```
eigen_values <- eigen(r)$values
eigen_values

plot(eigen_values, type="b", main="Scree Plot for Factor Analysis",
     xlab="Factor Number", ylab="Eigenvalue")
abline(h=1, col="red")

fa_result <- factanal(data_new, factors = 3, rotation = "varimax")

fa_result
fa_result$loadings
```

Output:



```
Call:
factanal(x = data_new, factors = 3, rotation = "varimax")
```

```
Uniquenesses:
      CF      SL      SU      FU      FS      DN      Age      CV
0.689 0.980 0.032 0.073 0.005 0.486 0.360 0.005
```

```
Loadings:
      Factor1 Factor2 Factor3
CF      0.545          0.115
SL      0.116
SU      0.974      0.126
FU      0.954      0.122
FS          0.997
DN      0.708
Age          0.793
CV      0.293      0.934 -0.191
```

```

      Factor1 Factor2 Factor3
SS loadings      2.761      1.922      0.687
Proportion Var      0.345      0.240      0.086
Cumulative Var      0.345      0.585      0.671
```

```
Test of the hypothesis that 3 factors are sufficient.
The chi square statistic is 1138.12 on 7 degrees of freedom.
The p-value is 1.69e-241
```

Comment:

Factor Analysis was conducted on the customer dataset using three factors and varimax rotation. The number of factors was selected based on the scree plot, where an elbow occurred at $i = 3$, indicating that three latent factors adequately summarize the correlation structure of the data. Variables such as Seconds of Use (SU), Frequency of Use (FU), Frequency of SMS (FS), and Customer Value (CV) have very low uniqueness values, meaning that these variables are well explained by the extracted factors. In contrast, Subscription Length (SL) has a very high uniqueness (0.980), suggesting that it is poorly explained by the factor model.

Interpretation of the factors:

Factor 1 is Voice Usage Intensity which has high loadings on Seconds of Use (SU = 0.974), Frequency of Use (FU = 0.954), Distinct Called Numbers (DN = 0.708), Call Failures (CF = 0.545), representing overall voice call usage intensity. Customers with high scores on this factor make frequent calls, spend more time on calls, contact many unique numbers, and experience a higher number of call failures due to intensive usage. This factor captures how heavily a customer uses voice call services.

Factor 2 consists of Frequency of SMS (FS = 0.997), Customer Value (CV = 0.934) which is Messaging and Customer Value, reflecting messaging behavior and revenue contribution. Customers scoring high on this factor send a high amount of SMS messages and produce higher customer value. This factor can be interpreted as a high-value, messaging-oriented customer segment.

Factor 3 primarily captures age-related characteristics of customers. Since Age loads strongly and independently on this factor so it represents a demographic dimension that is largely separate from usage and messaging behavior.

The three extracted factors together explain 67.1% of the total variance, indicating that a substantial amount of information in the original variables is retained using these latent dimensions.

3 No. Q. Ans.

Code:

```
library(MASS)
library(caret)
data <- read.csv(file.choose())
str(data)

colnames(data) <- c(
  "CF",
  "SL",
  "SU",
  "FU",
  "FS",
  "DN",
  "Age",
  "CV",
  "Churn"
)

head(data)

set.seed(111)
obs <- sample(1:nrow(data), round(0.7 * nrow(data)))

train.data <- data[obs, ]
test.data <- data[-obs, ]

train.data[,1:8] <- scale(train.data[,1:8])
test.data[,1:8] <- scale(test.data[,1:8])

lda.model <- lda(Churn ~ ., data = train.data)
lda.model$scaling

predictions <- predict(lda.model, test.data)$class
```

```
conf <- table(predicted = predictions, observed = test.data$Churn)
conf
```

```
confusionMatrix(conf)
```

Output:

```
> lda.model$scaling
      LD1
CF   0.68388572
SL  -0.02970837
SU  -0.31877167
FU  -0.81090127
FS  -1.13210558
DN  -0.40327639
Age  0.02774575
CV   0.78968389
~ |

Confusion Matrix and Statistics

      observed
predicted  0   1
      0  784 131
      1   11  19

      Accuracy : 0.8497
      95% CI   : (0.8253, 0.8719)
      No Information Rate : 0.8413
      P-Value [Acc > NIR] : 0.254

      Kappa : 0.167

      Mcnemar's Test P-Value : <2e-16

      Sensitivity : 0.9862
      Specificity : 0.1267
      Pos Pred Value : 0.8568
      Neg Pred Value : 0.6333
      Prevalence : 0.8413
      Detection Rate : 0.8296
      Detection Prevalence : 0.9683
      Balanced Accuracy : 0.5564

      'Positive' class : 0
```

Comment:

Linear Discriminant Analysis (LDA) classifies customers by constructing a linear discriminant function, which is a weighted linear combination of the predictor variables:

$$D = a_1(CF) + a_2(SL) + a_3(SU) + a_4(FU) + a_5(FS) + a_6(DN) + a_7(Age) + a_8(CV) + c$$

$$= 0.684(CF) - 0.030(SL) - 0.319(SU) - 0.811(FU) - 1.132(FS) - 0.403(DN) + 0.028(Age) + 0.790(CV)$$

Each customer receives a discriminant score, and classification is made by comparing this score to a cutoff value like: Customers closer to the non-churner centroid ($\text{Churn} = 0$) are classified as non-churners and customers closer to the churner centroid ($\text{Churn} = 1$) are classified as churners.

In our dataset, LDA achieved approximately 85% overall accuracy, 98.62% accuracy in non-churn detection and 12.67% accuracy in churn detection. The model is highly effective at identifying non-churners which means correctly classifying nearly all customers who remain with the company. But, the model has low ability to detect churners. This imbalance occurs because the dataset is heavily skewed toward non-churners.

The variables that contribute most to distinguishing between churners and non-churners can be identified by looking at the magnitude of the discriminant coefficients. Variables with larger absolute coefficients play a stronger role in separating the two groups. Such variables are Frequency of SMS (FS), Frequency of Use (FU), Customer Value (CV) and Call Failures (CF). Frequency of SMS (FS) has the largest discriminant coefficient, which means it is the most important variable in separating churners from non-churners. Customers who send SMS more frequently are more likely to stay with the company, while those with low SMS usage are more likely to churn. Frequency of Use (FU) is also an important factor. Customers who make calls more often tend to be non-churners, whereas customers with low call frequency are more likely to leave the service. Customer Value (CV) plays a strong role as well. Customers who generate higher revenue are more likely to remain loyal, while low-value customers are more prone to churn. Finally, Call Failures (CF) also contribute to group separation. Customers who experience more call failures are more likely to churn, showing that service quality issues increase the chance of leaving.